Rana Mozumder

Biomedical Engineer | Neuroscientist | NeuroAI Researcher

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Research Profile

Neuroscientist and biomedical engineer investigating how neural circuits encode, maintain, and transform information during cognition. My research integrates large-scale electrophysiology, optogenetics, computational modeling, and machine learning to understand neural population dynamics and develop technologies for neural interfacing and brain-computer interfaces.

Education

Vanderbilt University — **Nashville, TN, USA**
Ph.D. in Biomedical Engineering — *Aug 2022 - Expected Nov 2026*

- Advisor: Christos Constantinidis, Ph.D.
- Research: Cognitive neuroscience, NeuroAI, and Neuroengineering.
- Key Courses: Special Topics in Deep Learning, Advanced Quantitative Image Analysis, Analysis of fMRI, Quantitative Methods in BME, Fundamentals of Neuroscience II.

Bangladesh University of Engineering and Technology (BUET) — **Dhaka, Bangladesh**
B.Sc. in Biomedical Engineering — *2016 – 2021*

- CGPA: 3.78/4.00.
- Research: Biomedical signal processing and brain-computer interfaces.
- Key Courses: Digital Signal Processing, Random Signal Processing, Medical Imaging, Biomedical Instrumentation.

Research Interests

• Neural circuit computation • Computational neuroscience • Brain–computer interfaces • Electrophysiology • Neural population dynamics • Computational modeling • Neurotechnology

Research Experience

Vanderbilt University – Constantinidis Lab — **Nashville, TN, USA**
Graduate Research Assistant, Department of Biomedical Engineering — *Aug 2022 – Present*

- Investigate the neural circuit mechanisms underlying working memory and decision-making using large-scale electrophysiological recordings, cell-type-specific optogenetics, and computational analysis.
- Design and perform chronic in vivo electrophysiological recordings in behaving non-human primates using Neuropixels, laminar probes, Plexon, and custom microelectrode systems.
- Develop computational pipelines for neural population analysis, including spike sorting, population decoding, dimensionality reduction, statistical modeling, and characterization of neural dynamics.
- Develop recurrent neural network and dynamical-systems models to understand how population activity gives rise to behavior and compare model predictions with experimental neural recordings.
- Integrate electrophysiological, behavioral, and anatomical datasets, including CT/MRI registration for electrode localization and experimental planning.

Bangladesh University of Engineering and Technology — **Dhaka, Bangladesh**
Undergraduate Research Student — *2018 – 2021*

- Developed machine-learning and signal-processing methods for EEG-based mental-state and motor-imagery classification.
- Applied frequency-domain analysis, entropy-based features, variational mode decomposition, and machine-learning classifiers to biomedical signals.

Selected Projects

Optotagging Somatostatin Interneurons in Macaque Prefrontal Cortex *2024 – Present*
Vanderbilt University – Neuroengineering / Computational Neuroscience

- Develop and apply cell-type-specific optogenetic approaches to identify and manipulate somatostatin interneurons in dorsolateral prefrontal cortex of behaving macaques, combining optical stimulation with high-density electrophysiological recordings.
- Characterize cell-type-specific neural dynamics and their contributions to working memory using population decoding, dimensionality reduction, and statistical analysis of large-scale neural and behavioral datasets.
- Developing recurrent neural network and dynamical-systems models to investigate how population activity and temporal dynamics support working memory, and compare model dynamics with experimentally observed neural activity.

Asynchronous Firing and Off States in Working Memory Maintenance *2024-2025*
Computational Neuroscience / Neural Dynamics

- Investigated whether persistent activity during working memory reflects genuine single-trial neural dynamics or trial-averaging artifacts.
- Combined single-neuron analyses, inter-spike interval distributions, population decoding, and machine-learning classifiers to characterize sparse and asynchronous neural representations.

Behavioral Decoding Using Recurrent Neural Networks *2024*
Neuromatch – Brain-Inspired AI / Dynamical Systems

- Compared recurrent neural network architectures for decoding behavioral variables from neural population activity.
- Investigated how temporal history and distributed neural representations contribute to decision-making.

Latent Representation Learning of Neural Population Activity *2023*
Vanderbilt University – Representation Learning / Deep Learning

- Developed a transformer-based autoencoder to learn low-dimensional representations of neural activity during tasks.
- Visualized learned latent representations and evaluated their ability to distinguish correct and error trials.

fMRI Analysis of Visual Word Processing *2023*
Vanderbilt University – Biomedical Imaging / Neuroimaging

- Preprocessed fMRI data, performed GLM and connectivity analyses to characterize stimulus-related brain activity.
- Investigated differences in responses between the visual word form area and other cortical regions.

Deep Learning-Based Decoding of Motor Imagery *2021*
BUET – Brain-Computer Interfaces / Deep Learning

- Developed an ENET-Conv architecture using transfer learning for cross-dataset motor-imagery EEG decoding.
- Investigated methods for adapting BCI models across datasets with different EEG acquisition configurations.

Publications

[1] Woods, D.P., Adams, G.M., **Mozumder, R.**, Dang, W., Chen, A.Y., Chevée, M., Constantinidis, C., & Gonzales, D.L. "Repeatable, low-drift recordings in behaving non-human primates using flexible microelectrodes." ***Nature Communications***, 2026. In press.

[2] **Mozumder, R.**, Wang, Z., Dang, W., Zhu, J., Hammond, B.M., Machado, A., & Constantinidis, C. "Asynchronous firing and off states in working memory maintenance." ***Cell Reports***, 45(1), 2026.

[3] Karimi, A., **Mozumder, R.**, Schoenhaut, A.M., Rausis, O.G., Wallace, M.T., Ramachandran, R., & Constantinidis, C. "Integration of audiovisual motion in dorsolateral prefrontal cortical neurons." ***Journal of Neurophysiology***, 134(4), 1097–1110, 2025.

[4] **Mozumder, R.**, Chung, S., Li, S., & Constantinidis, C. "Contributions of narrow- and broad-spiking prefrontal and parietal neurons on working memory tasks." ***Frontiers in Systems Neuroscience***, 18, 1365622, 2024.

[5] **Mozumder, R.** & Constantinidis, C. "Single-neuron and population measures of neuronal activity in working memory tasks." ***Journal of Neurophysiology***, 130(3), 694–705, 2023.

Technical Skills

Programming	Python, MATLAB, R, C, Git, SQL
Machine Learning	Scikit-learn, NumPy, SciPy, Pandas, classification, decoding, clustering, statistical modeling
Deep Learning	PyTorch, TensorFlow, RNNs, Transformers autoencoders, representation learning
Neural Data Analysis	Spike sorting, PSTH analysis, population decoding, dimensionality reduction, neural dynamics, dynamical systems
Neuroimaging	fMRI, GLM, functional connectivity, CT/MRI registration and segmentation, AFNI, FSL, 3D Slicer
Experimental Methods	Optogenetics, Neuropixels, laminar electrophysiology, eye tracking, behavioral recording
Scientific Computing	MATLAB/Simulink, data visualization, exploratory data analysis, LaTeX

Teaching & Mentoring

Vanderbilt Summer Academy: Program for Talented Youth — **Nashville, TN, USA**
Instructor – The Thinking Brain: Unlocking Memory, Decisions, and the Prefrontal Cortex — *June 2026*

- Designed and taught a neuroscience course covering brain function, working memory, decision-making, and neurotechnology.
- Developed hands-on MATLAB activities for neural data analysis, visualization, and scientific computing.
- Mentored student teams on projects involving experimental design, brain-computer interfaces, and optogenetics.

Department of Biomedical Engineering, Vanderbilt University — **Nashville, TN, USA**
Graduate Teaching Assistant – Biomedical Instrumentation II — *Fall 2023, Fall 2024*

- Guided students in designing hardware and software systems for acquisition and analysis of physiological signals.
- Led laboratory sessions and mentored students in microcontroller programming, signal processing, troubleshooting, and experimental design.

Selected Presentations

Jun 2026	**Optogenetics in Non-Human Primates** – Session Instructor, MaDeLaNe Workshop
Apr 2026	**Effects of Activating Somatostatin Interneurons in Primate Prefrontal Cortex** – VIRAL Symposium, Talk
Nov 2025	**Optotagging SST Interneurons in Macaque Prefrontal Cortex Reveals Heterogeneous Physiology and Task Modulation** – Society for Neuroscience, Poster
Sep 2025	**Optotagging SST Interneurons in Macaque Prefrontal Cortex** – VBI Annual Retreat, Poster

Awards & Honors

Engineering Graduate Fellowship, Vanderbilt University School of Engineering	2022–2027
MedTech Design Sprint by Nissha Medical Technologies, Runner-up	Aug 2026
Travel Grant, Vanderbilt University School of Engineering	2026
Design for Life Competition – 4th Place	2020
Dean's List Honor, Bangladesh University of Engineering and Technology	2017, 2018, 2020

Leadership & Voluntary Experiences

Brain Blast by Vanderbilt Brain Institute, Volunteer	Mar 2026
Multi-Area, high-Density, Laminar Neurophysiology (MaDeLaNe) Workshop, Tutor	2025, 2026
Biomedical Engineering Graduate Student Association, Treasurer	2023-24

References

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